AMS 580

Unsupervised Learning: Cluster Analysis and Principal Component Analysis; Example

Name:________________________________SBU ID:_________________________________

Please include: (1) RMD code; (2) Output from RMD; (3) Answers to all the questions asked.

***Unsupervised Learning* with the Iris Data**

The ***iris*** data set, a built-in R data set, was introduced by the British statistician and biologist Ronald A. Fisher in his 1936 paper entitled "The use of multiple measurements in taxonomic problems". It is also referred to as Anderson's Iris data set because Edgar Anderson collected the data to quantify the morphologic variation of Iris flowers of three related species. The data set consists of 50 samples from each of three species of Iris (Iris Setosa, Iris virginica, and Iris versicolor). This sums to 150 records under 5 attributes - Petal Length, Petal Width, Sepal Length, Sepal width and Class (Species).

1. Our goal is to predict the iris Species based on the other 4 attributes given. Subsequently we wish to compare our data-driven clustering to that of the true species classification. For each of the following six clustering methods (K-mean; Hierarchical: Ward, Single-linkage, Complete-linkage, Average-linkage, Centroid), we need you to:

a) Perform the clustering analysis;
b) Draw scree-plot to see whether three clusters is reasonable or not;
c) Show the 2D representation of the Cluster solution;
d) Build a confusion matrix to evaluate the clustering performance;
    For Hierarchical clustering, please also draw the dendrogram showing the 3 clusters obtained;
e) Build a confusion matrix to compare the clustering results of the K-means and the Ward's method;
f) Finally, make a comparison of all six clustering methods, and rank their performance for the given problem. Now imagine you are meeting with a biologist, would you recommend the cluster analysis as an effective way to classify future unknown new flower species based on your analysis of the iris data? If your answer were yes, which method(s) would you recommend?

2. Please write up the entire RMD code necessary to answer the following questions use the same data set '**iris**':

a) Please compute the Principal Components (PC's) using the four quantitative attributes and print out a summary of the analysis – and in particular, please point out what percentage of the variations each PC would explain.
b) Please make a biplot, which includes both the position of each sample in terms of PC1 and PC2 and will also show you how the initial variables map onto this.
c) Now you will utilize the Species information binning the iris flowers into three groups: Iris Setosa, Iris virginica, and Iris versicolor. You will visualize the biplot by setting the ellipse argument to be TRUE, which will draw an ellipse around each group.

d) Last but not the least, we shall print out the PC1 as linear combinations of the original variables.
e) Now imagine you are meeting with a biologist, would you recommend the principal component analysis method as an effective way to classify future unknown new flower species based on your analysis of the iris data? Please let the scientist know why your answer is yes or no.

SCHULZ, C. 1954. *Peanuts.* United Feature Syndicate